



Divyang Deepak Palshetkar

Dual Degree (B.Tech-M.Tech)

Indian Institute of Technology Jodhpur

D21BCI001 [Dept. Rank: 1]

+91-9359849585

palshetkar.1@iitj.ac.in

palshetkardivyang@gmail.com

Portfolio GitHub LinkedIn

EDUCATION

Degree	Institute	Board / University	CGPA/Percentage	Year
B.Tech-M.Tech	Indian Institute of Technology Jodhpur	IIT JODHPUR	8.24	2026
Senior Secondary	Jaipuria School, Navi Mumbai	CBSE	94.6%	2020

SKILLS

- Frontend:** React.js, Next.js, Redux, HTML5, CSS3, JavaScript, TypeScript, Tailwind CSS, HTML Canvas, Figma
- Backend & APIs:** Node.js, Firebase, Express.js, REST APIs, CI/CD Workflows, Agile (Scrum), MongoDB, DevOps, AWS
- Programming & Algorithms:** Python, R, SQL, DSA, JavaScript, C++, SAS, Timeseries Analysis
- AI / ML:** Generative AI & Multimodal AI Agents, TensorFlow.js, NLP, Transformers, LSTM, LangChain, Scikit-learn
- Data Engineering & Analytics:** Data Pipelines, Advanced Excel, DAX, PowerBI, GCP (BigQuery, Cloud Storage), ETL, Data Modeling, Data Warehousing (MSSQL, PostgreSQL), Pandas, NumPy, Matplotlib, Power BI, Data Quality & Governance

WORK EXPERIENCE

- AI Research & Data Analysis Internship** May 2024 – July 2024
AICOE Project, Ministry of Education | Certificate IIT Jodhpur
 - Built a **large-scale analytics** and computer vision pipeline for more than 150 hours of traffic footage from 16 smart-city locations. The system automatically extracted key metrics including vehicle counts, speeds and congestion patterns.
 - Used **statistical modelling**, clustering, and anomaly detection to uncover patterns in urban traffic behaviour and support data-driven mobility planning.
 - Explored **generative AI** and **agent-based digital twins** to simulate traffic flow for predictive infrastructure planning.
- Jal Jeevan Mission, Ministry of Jal Shakti** May 2024 - July 2024
Internship | Certificate Rajasthan
 - Conducted **water surveys** and mapped 40+ dry borewells using **GPS, QGIS, and Global Mapper** to optimize groundwater recharge. Ensured compliance with environmental standards for sustainable water resource management.
 - Designed **3D filtration models** and horizontal recharge techniques, improving groundwater replenishment efficiency by 30%
 - Developed **REST APIs** to integrate GIS data with an analytics dashboard.

PROJECTS

- AI Market Intelligence Platform — Multi-Agent Financial Analytics System** October 2025
Python, Multi-Agent AI, LLM, Finance, Statsmodels, Plotly Live Demo | Video Exp. | Github
 - Built an **agentic financial analysis system** with **6 AI agents** (market, sentiment, fundamentals, risk, portfolio, orchestration) and a chat interface delivering insights and automated reports within seconds.
 - Designed a global data pipeline covering **500+ tickers** and more than a decade of price, fundamental and news data.
 - Expanded analytics with **rolling volatility, beta, drawdowns, VaR/CVaR, stress testing, LLM-based chart explain**, and **5,000+** Efficient Frontier simulations, enabling faster, benchmark-aware decision-making.
- Spotify Streaming Analytics Dashboard** October 2025
Power BI, SQL, DAX
 - Built a full Power BI dashboard analyzing streaming behaviour, viewer retention, artist performance, and popularity trends.
 - Used SQL-based transformations and DAX measures for time-series analysis, segmentation, and KPI computation.
 - Designed a data storytelling structure enabling clear business insights into global streaming trends.
- Data Warehouse & Analytics Project** August 2025
MS-SQL Server, SSMS, ETL, Data Modeling, BI Tools Github
 - Designed and developed an end-to-end data warehousing solution using the **Medallion Architecture (Bronze-Silver-Gold layers)** to transform raw **ERP and CRM datasets** into analytics-ready models.
 - Built **ETL pipelines** in **SQL Server** to automate **data extraction, cleansing, and integration** from multiple **CSV sources**, improving data reliability and consistency. Modeled fact and dimension tables in a star schema, optimizing query performance and supporting advanced **business analytics**.
 - Designed analytical data models that improved business reporting, KPI tracking, and query performance.
- Swiggy - Data Engineering Pipeline Project** July 2025
Python, SQL, Snowflake, ETL, Medallion Architecture, Data Quality, Orchestration
 - Architected an end-to-end data-wrangling pipeline for **large datasets**, implementing the **Medallion Architecture** to ingest, cleanse, transform, and model transactional data into analytics-ready fact and dimension tables using **SQL** and **Python-based ETL** scripts.
 - Integrated **data validation, schema enforcement, audit logging, incremental loads, and change tracking**, ensuring data quality, lineage, and governance.
 - Automated the pipeline using **task orchestration and scheduling**, enabling **real-time** refresh of **business dashboards** and KPIs for **order volumes, revenue trends, and delivery efficiency**, reducing manual effort and **pipeline latency** by over **80%**.
- Inventory Optimization & Sales Analytics | Slooze Take-Home Challenge** July 2025
Python, Pandas, Time-Series Analysis, Data Visualization Github
 - Developed an **end-to-end pipeline** to perform **demand forecasting, inventory optimization** (EOQ, reorder points), and **ABC classification** of SKUs.

- Conducted **EDA** on sales, purchase, and lead time data to uncover **seasonal trends**, **top-performing products**, and insights.
- Construction Project Monitoring & Reporting System** May 2025
Advanced Excel, Business, Management
 - Built a multi-sheet Excel-based analytics system integrating DPR, WBS, manpower, material, QC, safety, equipment, billing data, automated Master Schedule, Look-Ahead Plans, weekly progress summaries, and Earned Value tracking.
 - Created dashboards for planned vs. actual progress, cost variances, material trends, manpower utilization, and incident patterns.
- ExpensIQ – AI-Powered Personal Finance Platform** October 2025
Full-Stack Project | Next.js, TypeScript, Prisma ORM, Supabase, Inngest, Arcjet Live Demo | Github
 - Developed a full-stack, **AI-powered personal finance platform** with features like **income/expense tracking**, **multi-account budgeting**, **receipt scanning** using Gemini AI + OCR, and monthly summaries with visualizations and **AI insights**
 - Implemented **recurring transaction automation**, **real-time budget alerts** via Resend + Inngest, secure Google authentication using Clerk and bot protection with Arcjet.
- PulseMeet - Telemedicine Appointment App** August 2025
Full-Stack Project | Next.js, Prisma ORM, Vonage Video API, PostgreSQL (Neon) Live Demo | Github
 - Built **PulseMeet**, a **full-stack telemedicine platform** enabling patients to consult doctors via **secure 1:1 video calls** (Vonage), manage **real-time appointment scheduling**, and subscribe to **credit-based plans** using Clerk + Stripe.
 - Integrated **role-based access control** (Patients, Doctors, Admins), **Clerk authentication**, **Vonage encrypted video sessions**, **credit tracking**, **subscription management**, and a **payout system** for doctors.
- CookSmartAI – AI For Kitchen & Diet** August 2025
Full-Stack Project | Next.js, Prisma ORM, Supabase, Hugging Face Live Demo | Github
 - Built a **full-stack, AI-powered culinary platform** enabling **recipe generation**, **custom meal plans**, **nutritional analysis**, and **auto-generated dish images** using **Gemini AI**, **Hugging Face**, and **Pollination AI**.
 - Implemented secure **Clerk authentication**, real-time **PostgreSQL database** via **Supabase + Prisma**, responsive **Next.js & Tailwind CSS UI**, with features like **AI Image Scanner**, **global recipe hub** with advanced filters, and **shopping list generation**.
- Multimodal AI Travel Planning Agent** July 2025
Python, LangChain, Llama, Geopy, Folium Github
 - Engineered a **multimodal AI agent** that autonomously generates personalized, day-wise travel itineraries by combining **LLM reasoning** (Groq-Llama-3.3) with geospatial mapping, weather data, and user preferences.
 - Utilized **LangChain** for prompt orchestration, structured JSON generation, and conversational planning, enabling the agent to plan realistic, route-optimized travel experiences.
- iPad Math Clone (DrawAI)** June 2025
Full-Stack Project | React.js, Next.js, Canvas, TensorFlow.js, Math.js, REST API Live Demo | Github
 - Developed a **React-based interactive Canvas application** that mimics iPad-style math note-taking, enabling users to **draw**, **sketch**, and **visualize** mathematical diagrams with **real-time AI analysis** using Google Gemini API.
 - Implemented **advanced Canvas rendering** logic with shape tools, color palette, eraser, grid/ruler systems, and **stroke tracking** to deliver precise, high-performance drawing interactions similar to professional visualization tools.
 - Built a modular and scalable UI using **Tailwind CSS**, integrated **state management**, **analytics dashboard**(similar to **PowerBI**), and integrated AI-assisted math recognition using **TensorFlow.js** and **Math.js** for real-time expression parsing and visualization with **local storage** persistence for offline data saving.
- Role-Based Access Control (RBAC) System** June 2025
React.js, TypeScript, Express.js, MongoDB, React Router Github
 - Built a **scalable and responsive RBAC interface** with **dynamic user and role management**, including **fine-grained permission controls** and secure route-based access.
 - Designed **modular, maintainable components** following clean architecture principles, optimized with caching and persistent local storage for enhanced performance.
 - Implemented **role hierarchy** (Admin, Moderator, User) and integrated real-time permission updates, ensuring robust security and consistent UX across user tiers.
- Fake News Detection using TensorFlow** May 2025
Deep Learning Project | Python, NLP, TensorFlow Github
 - Developed a **Deep Learning-based Fake News Classifier** capable of distinguishing between **real** and **fake news** articles using textual and metadata features. Conducted **EDA** to identify author-wise and article-length patterns, visualizing insights using **Matplotlib** and **Seaborn**.
 - Implemented a complete **data preprocessing pipeline** and **label encoding** for categorical features. Designed and trained an **LSTM neural network** using **TensorFlow/Keras**, achieving an accuracy of approximately **94%**.

POSITIONS OF RESPONSIBILITY

- Festival Chief**,Ganeshotsav’24 IIT Jodhpur - Organized cultural events, leading a team of **200+** volunteers. 2024
- Head**,Public Relation Team, EDIFICIO 2023